

# Industrial CO2 Emission Prediction Report

Report Generated: 17-09-2026 13:13:12  
Record Timestamp: 17-09-2026 13:13:11  
Prediction Record ID: #59

## Core Telemetry Parameters

| Parameter              | Observed Value   |
|------------------------|------------------|
| Energy Consumption     | 14.20 kWh        |
| Production Count       | 115.00 Units     |
| Working / Shift Hour   | 15:00 hrs        |
| Manufacturing Schedule | Standard Weekday |

## Emission Forecast & Analysis

|                          |                                      |
|--------------------------|--------------------------------------|
| Forecasted CO2 Emission  | 13.07 kg CO2                         |
| Pollution Classification | Low                                  |
| Inference Methodology    | Random Forest Machine Learning Model |
| Grid Emission Constant   | 0.82 kg CO2/kWh                      |

## Recommended Mitigation Measures

- Continue monitoring industrial energy consumption.
- Maintain machines regularly for energy-efficient operation.
- Continue using high-efficiency motors and variable frequency drives.
- Monitor carbon emissions periodically against baseline.

This certification report was generated by the Industrial Carbon Emission Forecasting System. Telemetry input is evaluated using Scikit-Learn Machine Learning models and Scope 2 emission standards.